Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Countyline LLC

Physical address of dairy:

15453 Monte Vista Ave	Denair	Stanislaus	95316
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 05/15/2002

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X020-X013-X027-XXXX X042-X010-X014-XXXX

B. OPERATORS

Ahlem, Vance William

Operator name: Ahlem, Vance William

Telephone no.: (209) 874-2477 (Landline) (Cellular)

19266 E Fowler AVE	Turlock	CA	95380
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Ahlem, Vance William

Legal owner name: Ahlem, Vance William

Telephone no.: (209) 874-2477 (Landline) (Cellular)

19266 E Fowler AVE	Turlock	CA	95380
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	250	0
Number under roof	1,926	286	112	0	0	0
Maximum number	1,926	286	112	0	250	0
Average number	1,926	286	112	0	250	0
Avg live weight (lbs)	950	1,050	650	0		

Predominant milk cow breed: Jersey

Average milk production: 55 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 51,621.73 *tons per reporting period*

Total nitrogen from manure: 677,210.32 *lbs per reporting period*

After ammonia losses (30% loss applied): 474,047.22 *lbs per reporting period*

Total phosphorus from manure: 110,794.77 *lbs per reporting period*

Total potassium from manure: 349,866.04 *lbs per reporting period*

Total salt from manure: 972,622.80 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 17,304,000 *gallons*

Total nitrogen generated: 139,844.11 *lbs*

Total phosphorus generated: 6,905.49 *lbs*

Total potassium generated: 161,747.63 *lbs*

Total salt generated: 811,468.47 *lbs*

	17,304,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	17,304,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW-Bledsoe PG&E	Ground water
IW-Bledsoe TID	Ground water
IW-Dairy TID	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
07/18/2023	Solid commercial fertilizer *UAN 32*	39.88 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	25,497.68	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	25,497.68	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Bledsoe Dryland	80	80	1	manure	X020-X013-X013-XXXX X042-X020-X017-XXXX
Dairy Dryland	50	50	1	manure	X020-X013-X037-XXXX X042-X010-X014-XXXX
Field 1	35	35	2	process wastewater	X020-X013-X037-XXXX
Field 2	10	10	2	process wastewater	X042-X010-X014-XXXX
Field 3	77	77	2	process wastewater	X042-X010-X014-XXXX
Field 4	35	10	2	process wastewater	X042-X010-X014-XXXX
Field 5	116	116	2	process wastewater	X042-X010-X014-XXXX
Field 6	30	30	2	manure	X042-X020-X017-XXXX
Field 8	130	130	2	manure	X042-X020-X017-XXXX
Field 9	159	159	2	manure	X042-X020-X017-XXXX
Totals for areas that were used for application	722	697	18		
Totals for areas that were not used for application					
Land application area totals	722	697	18		

B. CROPS AND HARVESTS

Bledsoe Dryland

Field name: Bledsoe Dryland

11/27/2022: Oats, hay

Crop: Oats, hay Acres planted: 80 Plant date: 11/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	266.00 *ton*	As-is		10.0	9,270.00	1,810.00	11,400.00		6.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	120.00	19.50	99.00	0.00
Total actual harvest content	3.33	61.65	12.04	75.81	397.40

Dairy Dryland

Dairy Dryland

Field name: Dairy Dryland

11/25/2022: Oats, hay

Crop: Oats, hay Acres planted: 50 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	185.00 *ton*	As-is		10.0	9,270.00	1,810.00	11,400.00		6.64

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	120.00	19.50	99.00	0.00
Total actual harvest content	3.70	68.60	13.39	84.36	442.22

Field 1

Field name: Field 1

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	522.00 *ton*	As-is		71.4	5,910.00	1,230.00	11,900.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.91	176.29	36.69	354.96	921.34

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/10/2023	868.40 *ton*	As-is		69.7	3,510.00	480.00	4,800.00		5.18

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	24.81	174.18	23.82	238.19	778.85

Field 2

Field name: Field 2

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	146.00 *ton*	As-is		72.2	6,290.00	1,770.00	16,500.00		13.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.60	183.67	51.68	481.80	1,055.29

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/10/2023	280.00 *ton*	As-is		70.2	4,140.00	730.00	6,740.00		7.15

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	28.00	231.84	40.88	377.44	1,193.19

Field 3

Field name: Field 3

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 77 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	861.00 *ton*	As-is		69.2	8,800.00	1,890.00	18,400.00		15.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	11.18	196.80	42.27	411.49	1,095.19

Field 3

05/18/2023: Corn, silage

Crop: Corn, silage Acres planted: 77 Plant date: 05/18/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	2,209.80 *ton*	As-is		68.6	5,370.00	680.00	8,590.00		8.34

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	28.70	308.22	39.03	493.04	1,503.10

Field 4

Field name: Field 4

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	142.00 *ton*	As-is		61.6	8,660.00	1,760.00	19,100.00		12.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	14.20	245.94	49.98	542.44	1,308.67

05/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 05/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	300.00 *ton*	As-is		70.0	4,530.00	660.00	7,790.00		7.22

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	30.00	271.80	39.60	467.40	1,299.60

Field 5

Field name: Field 5

11/25/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 116 Plant date: 11/25/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/22/2023	1,409.00 *ton*	As-is		64.0	7,850.00	1,820.00	19,000.00		14.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	14.00	140.00	22.40	116.20	0.00
Total actual harvest content	12.15	190.70	44.21	461.57	1,303.08

05/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 116 Plant date: 05/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	2,820.00 *ton*	As-is		68.5	4,880.00	630.00	8,180.00		7.68

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	24.31	237.27	30.63	397.72	1,176.23

Field 6

Field name: Field 6

11/26/2022: Oats, hay

Crop: Oats, hay Acres planted: 30 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	188.00 *ton*	As-is		19.4	8,060.00	1,820.00	11,000.00		6.72

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	5.00	200.00	32.50	165.00	0.00
Total actual harvest content	6.27	101.02	22.81	137.87	678.85

Field 6

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/20/2023	812.00 *ton*	As-is		60.6	4,650.00	830.00	3,600.00		2.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	27.07	251.72	44.93	194.88	490.56

Field 8

Field name: Field 8

11/26/2022: Oats, hay

Crop: Oats, hay Acres planted: 130 Plant date: 11/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	753.00 *ton*	As-is		10.4	8,630.00	1,880.00	10,800.00		6.04

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	5.00	200.00	32.50	165.00	0.00
Total actual harvest content	5.79	99.98	21.78	125.11	626.94

06/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 130 Plant date: 06/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	3,499.35 *ton*	As-is		65.8	4,860.00	1,070.00	5,350.00		9.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	26.92	261.64	57.60	288.02	1,657.08

Field 9

Field name: Field 9

11/27/2022: Oats, hay

Crop: Oats, hay Acres planted: 159 Plant date: 11/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/21/2023	876.00 *ton*	As-is		8.8	9,210.00	1,930.00	9,650.00		6.15

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	5.00	200.00	32.50	165.00	0.00
Total actual harvest content	5.51	101.48	21.27	106.33	618.03

06/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 159 Plant date: 06/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	4,658.45 *ton*	As-is		64.2	4,980.00	960.00	4,390.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	29.30	291.81	56.25	257.24	1,804.08

NUTRIENT BUDGET

A. LAND APPLICATIONS

Bledsoe Dryland - 11/27/2022: Oats, hay

Field name: Bledsoe Dryland

Crop: Oats, hay — Plant date: 11/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator	Separator solids	71.12	7.00	28.45	314.54	470.00 *ton*
Application event totals		71.12	7.00	28.45	314.54	

Dairy Dryland - 11/25/2022: Oats, hay

Field name: Dairy Dryland

Crop: Oats, hay — Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator	Separator solids	77.48	7.63	30.99	342.65	320.00 *ton*
Application event totals		77.48	7.63	30.99	342.65	

Field 1 - 11/25/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 1 - 11/25/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.62	6.14	121.37	627.28	504,000.00 *gal*
IW-Dairy TID	Ground water	1.40	0.00	0.00	113.30	3,600,000.00 *gal*
Application event totals		70.02	6.14	121.37	740.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	165.43	7.84	165.43	1,029.04	588,000.00 *gal*
IW-Dairy TID	Ground water	1.40	0.00	0.00	113.30	3,600,000.00 *gal*
Application event totals		166.83	7.84	165.43	1,142.34	

Field 1 - 05/18/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.72	3.92	82.72	514.52	294,000.00 *gal*
IW-Dairy TID	Ground water	1.40	0.00	0.00	113.30	3,600,000.00 *gal*
Application event totals		84.11	3.92	82.72	627.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.31	0.00	0.00	105.75	3,360,000.00 *gal*
Application event totals		1.31	0.00	0.00	105.75	

Field 1 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.21	0.00	0.00	98.19	3,120,000.00 *gal*
Application event totals		1.21	0.00	0.00	98.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.12	0.00	0.00	90.64	2,880,000.00 *gal*
Application event totals		1.12	0.00	0.00	90.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.12	0.00	0.00	90.64	2,880,000.00 *gal*
Application event totals		1.12	0.00	0.00	90.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.80	2.43	79.21	259.36	294,000.00 *gal*
IW-Dairy TID	Ground water	1.07	0.00	0.00	86.86	2,760,000.00 *gal*
Application event totals		71.87	2.43	79.21	346.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.03	0.00	0.00	83.09	2,640,000.00 *gal*
Application event totals		1.03	0.00	0.00	83.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	0.98	0.00	0.00	79.31	2,520,000.00 *gal*
Application event totals		0.98	0.00	0.00	79.31	

Field 1 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.80	2.43	79.21	259.36	294,000.00 *gal*
IW-Dairy TID	Ground water	0.93	0.00	0.00	75.53	2,400,000.00 *gal*
Application event totals		71.73	2.43	79.21	334.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	0.84	0.00	0.00	67.98	2,160,000.00 *gal*
Application event totals		0.84	0.00	0.00	67.98	

Field 2 - 11/25/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.05	7.16	141.60	731.82	168,000.00 *gal*
IW-Dairy TID	Ground water	1.63	0.00	0.00	132.18	1,200,000.00 *gal*
Application event totals		81.68	7.16	141.60	864.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	165.43	7.84	165.43	1,029.04	168,000.00 *gal*
IW-Dairy TID	Ground water	1.63	0.00	0.00	132.18	1,200,000.00 *gal*
Application event totals		167.06	7.84	165.43	1,161.22	

Field 2 - 05/18/2023: Corn, silage

Field 2 - 05/18/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	124.07	5.88	124.07	771.78	126,000.00 *gal*
IW-Dairy TID	Ground water	1.35	0.00	0.00	109.05	990,000.00 *gal*
Application event totals		125.42	5.88	124.07	880.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.18	0.00	0.00	95.83	870,000.00 *gal*
Application event totals		1.18	0.00	0.00	95.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.18	0.00	0.00	95.83	870,000.00 *gal*
Application event totals		1.18	0.00	0.00	95.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.14	0.00	0.00	92.53	840,000.00 *gal*
Application event totals		1.14	0.00	0.00	92.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.14	0.00	0.00	92.53	840,000.00 *gal*
Application event totals		1.14	0.00	0.00	92.53	

Field 2 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.20	3.65	118.82	389.04	126,000.00 *gal*
IW-Dairy TID	Ground water	0.98	0.00	0.00	79.31	720,000.00 *gal*
Application event totals		107.18	3.65	118.82	468.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.06	0.00	0.00	85.92	780,000.00 *gal*
Application event totals		1.06	0.00	0.00	85.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	0.98	0.00	0.00	79.31	720,000.00 *gal*
Application event totals		0.98	0.00	0.00	79.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.80	2.43	79.21	259.36	84,000.00 *gal*
IW-Dairy TID	Ground water	0.98	0.00	0.00	79.31	720,000.00 *gal*
Application event totals		71.78	2.43	79.21	338.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	0.98	0.00	0.00	79.31	720,000.00 *gal*
Application event totals		0.98	0.00	0.00	79.31	

Field 3 - 11/25/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 11/25/2022

Field 3 - 11/25/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.57	8.37	165.50	855.38	1,512,000.00 *gal*
IW-Dairy TID	Ground water	1.91	0.00	0.00	154.50	10,800,000.00 *gal*
Application event totals		95.48	8.37	165.50	1,009.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	161.13	7.63	161.13	1,002.31	1,260,000.00 *gal*
IW-Dairy TID	Ground water	1.85	0.00	0.00	150.21	10,500,000.00 *gal*
Application event totals		162.99	7.63	161.13	1,152.52	

Field 3 - 05/18/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 05/18/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	128.91	6.11	128.91	801.85	1,008,000.00 *gal*
IW-Dairy TID	Ground water	1.51	0.00	0.00	121.88	8,520,000.00 *gal*
Application event totals		130.41	6.11	128.91	923.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.44	0.00	0.00	116.73	8,160,000.00 *gal*
Application event totals		1.44	0.00	0.00	116.73	

Field 3 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.44	0.00	0.00	116.73	8,160,000.00 *gal*
Application event totals		1.44	0.00	0.00	116.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.46	0.00	0.00	118.45	8,280,000.00 *gal*
Application event totals		1.46	0.00	0.00	118.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.46	0.00	0.00	118.45	8,280,000.00 *gal*
Application event totals		1.46	0.00	0.00	118.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.34	3.79	123.45	404.20	1,008,000.00 *gal*
IW-Dairy TID	Ground water	1.44	0.00	0.00	116.73	8,160,000.00 *gal*
Application event totals		111.78	3.79	123.45	520.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.40	0.00	0.00	113.30	7,920,000.00 *gal*
Application event totals		1.40	0.00	0.00	113.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.40	0.00	0.00	113.30	7,920,000.00 *gal*
Application event totals		1.40	0.00	0.00	113.30	

Field 3 - 05/18/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	110.34	3.79	123.45	404.20	1,008,000.00 *gal*
IW-Dairy TID	Ground water	1.38	0.00	0.00	111.58	7,800,000.00 *gal*
Application event totals		111.71	3.79	123.45	515.79	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.36	0.00	0.00	109.87	7,680,000.00 *gal*
Application event totals		1.36	0.00	0.00	109.87	

Field 4 - 11/25/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	120.08	10.75	212.40	1,097.73	252,000.00 *gal*
IW-Dairy TID	Ground water	2.45	0.00	0.00	198.28	1,800,000.00 *gal*
Application event totals		122.53	10.75	212.40	1,296.01	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	165.43	7.84	165.43	1,029.04	168,000.00 *gal*
IW-Dairy TID	Ground water	2.04	0.00	0.00	165.23	1,500,000.00 *gal*
Application event totals		167.47	7.84	165.43	1,194.27	

Field 4 - 05/19/2023: Corn, silage

Field 4 - 05/19/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 05/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.80	0.00	0.00	145.40	1,320,000.00 *gal*
Application event totals		1.80	0.00	0.00	145.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	124.07	5.88	124.07	771.78	126,000.00 *gal*
Application event totals		124.07	5.88	124.07	771.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.47	0.00	0.00	118.97	1,080,000.00 *gal*
Application event totals		1.47	0.00	0.00	118.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.39	0.00	0.00	112.36	1,020,000.00 *gal*
Application event totals		1.39	0.00	0.00	112.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.39	0.00	0.00	112.36	1,020,000.00 *gal*
Application event totals		1.39	0.00	0.00	112.36	

Field 4 - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.20	3.65	118.82	389.04	126,000.00 *gal*
IW-Dairy TID	Ground water	1.35	0.00	0.00	109.05	990,000.00 *gal*
Application event totals		107.55	3.65	118.82	498.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.31	0.00	0.00	105.75	960,000.00 *gal*
Application event totals		1.31	0.00	0.00	105.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.22	0.00	0.00	99.14	900,000.00 *gal*
Application event totals		1.22	0.00	0.00	99.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.20	3.65	118.82	389.04	126,000.00 *gal*
IW-Dairy TID	Ground water	1.22	0.00	0.00	99.14	900,000.00 *gal*
Application event totals		107.42	3.65	118.82	488.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.22	0.00	0.00	99.14	900,000.00 *gal*
Application event totals		1.22	0.00	0.00	99.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.22	0.00	0.00	99.14	900,000.00 *gal*
Application event totals		1.22	0.00	0.00	99.14	

Field 5 - 11/25/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough

Plant date: 11/25/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.81	7.41	146.48	757.06	2,016,000.00 *gal*
IW-Dairy TID	Ground water	1.69	0.00	0.00	136.74	14,400,000.00 *gal*
Application event totals		84.50	7.41	146.48	893.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	171.14	8.11	171.14	1,064.52	2,016,000.00 *gal*
IW-Dairy TID	Ground water	1.55	0.00	0.00	125.35	13,200,000.00 *gal*
Application event totals		172.68	8.11	171.14	1,189.87	

Field 5 - 05/19/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage

Plant date: 05/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	128.35	6.08	128.35	798.39	1,512,000.00 *gal*
IW-Dairy TID	Ground water	1.55	0.00	0.00	125.35	13,200,000.00 *gal*
Application event totals		129.90	6.08	128.35	923.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.41	0.00	0.00	113.95	12,000,000.00 *gal*
Application event totals		1.41	0.00	0.00	113.95	

Field 5 - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.37	0.00	0.00	111.10	11,700,000.00 *gal*
Application event totals		1.37	0.00	0.00	111.10	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.37	0.00	0.00	111.10	11,700,000.00 *gal*
Application event totals		1.37	0.00	0.00	111.10	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.30	0.00	0.00	105.41	11,100,000.00 *gal*
Application event totals		1.30	0.00	0.00	105.41	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.55	3.15	102.43	335.38	1,260,000.00 *gal*
IW-Dairy TID	Ground water	1.23	0.00	0.00	99.71	10,500,000.00 *gal*
Application event totals		92.78	3.15	102.43	435.09	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.27	0.00	0.00	102.56	10,800,000.00 *gal*
Application event totals		1.27	0.00	0.00	102.56	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.27	0.00	0.00	102.56	10,800,000.00 *gal*
Application event totals		1.27	0.00	0.00	102.56	

Field 5 - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	91.55	3.15	102.43	335.38	1,260,000.00 *gal*
IW-Dairy TID	Ground water	1.23	0.00	0.00	99.71	10,500,000.00 *gal*
Application event totals		92.78	3.15	102.43	435.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Dairy TID	Ground water	1.20	0.00	0.00	96.86	10,200,000.00 *gal*
Application event totals		1.20	0.00	0.00	96.86	

Field 6 - 11/26/2022: Oats, hay

Field name: Field 6

Crop: Oats, hay Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator	Separator solids	129.13	12.71	51.65	571.08	320.00 *ton*
Application event totals		129.13	12.71	51.65	571.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.58	0.00	0.00	133.52	3,750,000.00 *gal*
Application event totals		1.58	0.00	0.00	133.52	

Field 6 - 06/29/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage Plant date: 06/29/2023

Field 6 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.76	0.00	0.00	149.54	4,200,000.00 *gal*
Application event totals		1.76	0.00	0.00	149.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	264.32	91.19	2.64	5,127.81	480.00 *ton*
Application event totals		264.32	91.19	2.64	5,127.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.64	0.00	0.00	138.86	3,900,000.00 *gal*
Application event totals		1.64	0.00	0.00	138.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UAN 32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
IW-Bledsoe TID	Ground water	1.51	0.00	0.00	128.18	3,600,000.00 *gal*
Application event totals		81.51	0.00	0.00	128.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.39	0.00	0.00	117.50	3,300,000.00 *gal*
Application event totals		1.39	0.00	0.00	117.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.26	0.00	0.00	106.82	3,000,000.00 *gal*
Application event totals		1.26	0.00	0.00	106.82	

Field 6 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.13	0.00	0.00	96.13	2,700,000.00 *gal*
Application event totals		1.13	0.00	0.00	96.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.13	0.00	0.00	96.13	2,700,000.00 *gal*
Application event totals		1.13	0.00	0.00	96.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.13	0.00	0.00	96.13	2,700,000.00 *gal*
Application event totals		1.13	0.00	0.00	96.13	

Field 8 - 11/26/2022: Oats, hay

Field name: Field 8

Crop: Oats, hay

Plant date: 11/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/17/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator	Separator solids	130.37	12.84	52.15	576.58	1,400.00 *ton*
Application event totals		130.37	12.84	52.15	576.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.63	0.00	0.00	138.04	16,800,000.00 *gal*
Application event totals		1.63	0.00	0.00	138.04	

Field 8 - 06/29/2023: Corn, silage

Field name: Field 8

Crop: Corn, silage

Plant date: 06/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.54	0.00	0.00	130.64	15,900,000.00 *gal*
Application event totals		1.54	0.00	0.00	130.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	266.86	92.07	2.67	5,177.11	2,100.00 *ton*
Application event totals		266.86	92.07	2.67	5,177.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.48	0.00	0.00	125.71	15,300,000.00 *gal*
Application event totals		1.48	0.00	0.00	125.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UAN 32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
IW-Bledsoe TID	Ground water	1.40	0.00	0.00	118.32	14,400,000.00 *gal*
Application event totals		81.40	0.00	0.00	118.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.37	0.00	0.00	115.85	14,100,000.00 *gal*
Application event totals		1.37	0.00	0.00	115.85	

Field 8 - 06/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.34	0.00	0.00	113.39	13,800,000.00 *gal*
Application event totals		1.34	0.00	0.00	113.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.31	0.00	0.00	110.92	13,500,000.00 *gal*
Application event totals		1.31	0.00	0.00	110.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.31	0.00	0.00	110.92	13,500,000.00 *gal*
Application event totals		1.31	0.00	0.00	110.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe TID	Ground water	1.28	0.00	0.00	108.46	13,200,000.00 *gal*
Application event totals		1.28	0.00	0.00	108.46	

Field 9 - 11/27/2022: Oats, hay

Field name: Field 9

Crop: Oats, hay

Plant date: 11/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/18/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator	Separator solids	127.91	12.59	51.16	565.70	1,680.00 *ton*
Application event totals		127.91	12.59	51.16	565.70	

Field 9 - 11/27/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe PG&E	Ground water	1.42	0.00	0.00	133.83	20,400,000.00 *gal*
Application event totals		1.42	0.00	0.00	133.83	

Field 9 - 06/30/2023: Corn, silage

Field name: Field 9

Crop: Corn, silage

Plant date: 06/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral	Corral solids	270.14	93.20	2.70	5,240.68	2,600.00 *ton*
Application event totals		270.14	93.20	2.70	5,240.68	

06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe PG&E	Ground water	1.57	0.00	0.00	147.61	22,500,000.00 *gal*
Application event totals		1.57	0.00	0.00	147.61	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe PG&E	Ground water	1.42	0.00	0.00	133.83	20,400,000.00 *gal*
Application event totals		1.42	0.00	0.00	133.83	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UAN 32	Solid commercial fertilizer	80.00	0.00	0.00	0.00	
IW-Bledsoe PG&E	Ground water	1.37	0.00	0.00	128.91	19,650,000.00 *gal*
Application event totals		81.37	0.00	0.00	128.91	

Field 9 - 06/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe PG&E	Ground water	1.27	0.00	0.00	119.07	18,150,000.00 *gal*
Application event totals		1.27	0.00	0.00	119.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe PG&E	Ground water	1.24	0.00	0.00	116.12	17,700,000.00 *gal*
Application event totals		1.24	0.00	0.00	116.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe PG&E	Ground water	1.19	0.00	0.00	112.19	17,100,000.00 *gal*
Application event totals		1.19	0.00	0.00	112.19	

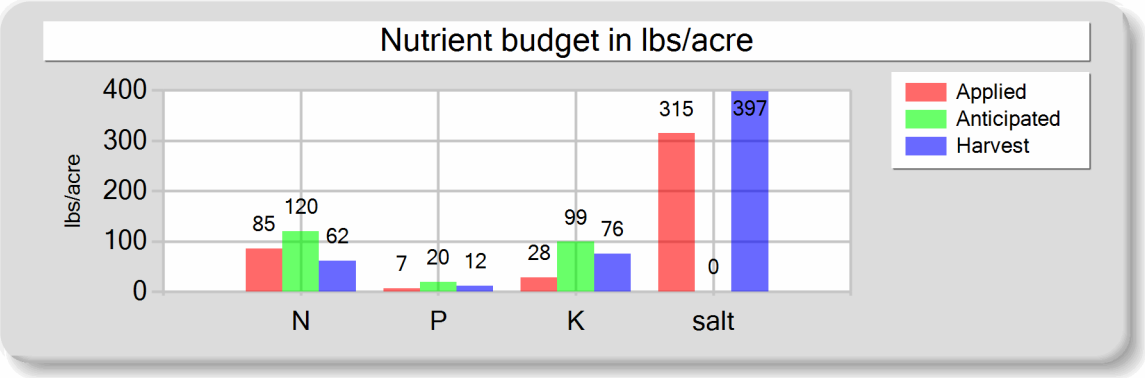
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW-Bledsoe PG&E	Ground water	1.11	0.00	0.00	104.31	15,900,000.00 *gal*
Application event totals		1.11	0.00	0.00	104.31	

B. NUTRIENT BUDGET

Bledsoe Dryland - 11/27/2022: Oats, hay

Field name: Bledsoe Dryland Crop: Oats, hay Plant date: 11/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	71.12	7.00	28.45	314.54
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	85.12	7.00	28.45	314.54
Anticipated crop nutrient removal	120.00	19.50	99.00	0.00
Actual crop nutrient removal	61.65	12.04	75.81	397.40
Nutrient balance	23.47	-5.03	-47.36	-82.86
Applied to removed ratio	1.38	0.58	0.38	0.79

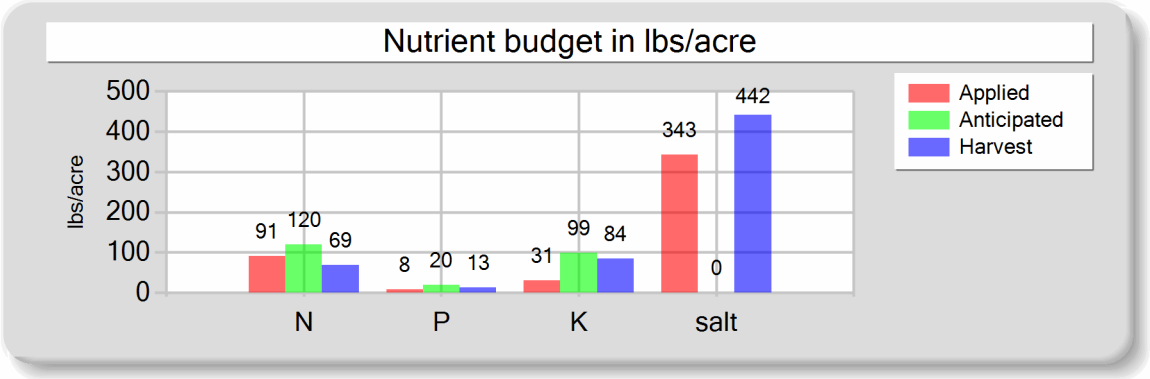
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Dairy Dryland - 11/25/2022: Oats, hay

Field name: Dairy Dryland Crop: Oats, hay Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	77.48	7.63	30.99	342.65
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	91.48	7.63	30.99	342.65
Anticipated crop nutrient removal	120.00	19.50	99.00	0.00
Actual crop nutrient removal	68.60	13.39	84.36	442.22
Nutrient balance	22.88	-5.77	-53.37	-99.57
Applied to removed ratio	1.33	0.57	0.37	0.77

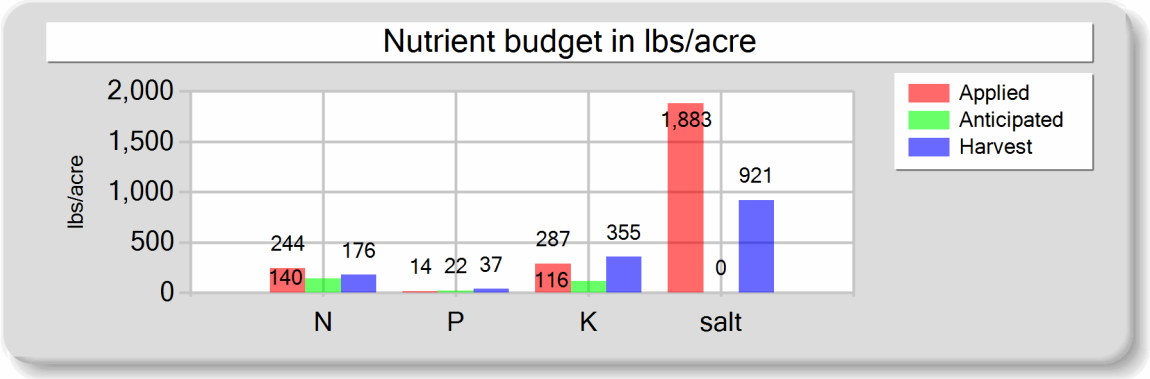
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 1 - 11/25/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	234.05	13.98	286.80	1,656.32
Fresh water	2.80	0.00	0.00	226.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	243.85	13.98	286.80	1,882.92
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	176.29	36.69	354.96	921.34
Nutrient balance	67.56	-22.71	-68.16	961.57
Applied to removed ratio	1.38	0.38	0.81	2.04

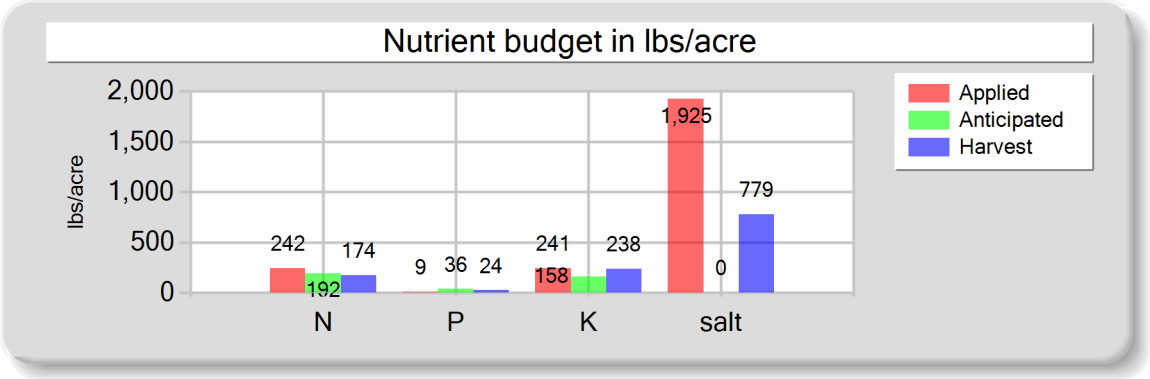
Fresh water applied	
7,200,000.00	*gallons*
265.15	*acre-inches*
7.58	*inches/acre*

Process wastewater applied	
1,092,000.00	*gallons*
40.21	*acre-inches*
1.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 1 - 05/18/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.31	8.78	241.14	1,033.24
Fresh water	11.01	0.00	0.00	891.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.32	8.78	241.14	1,924.55
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	174.18	23.82	238.19	778.85
Nutrient balance	68.14	-15.04	2.95	1,145.70
Applied to removed ratio	1.39	0.37	1.01	2.47

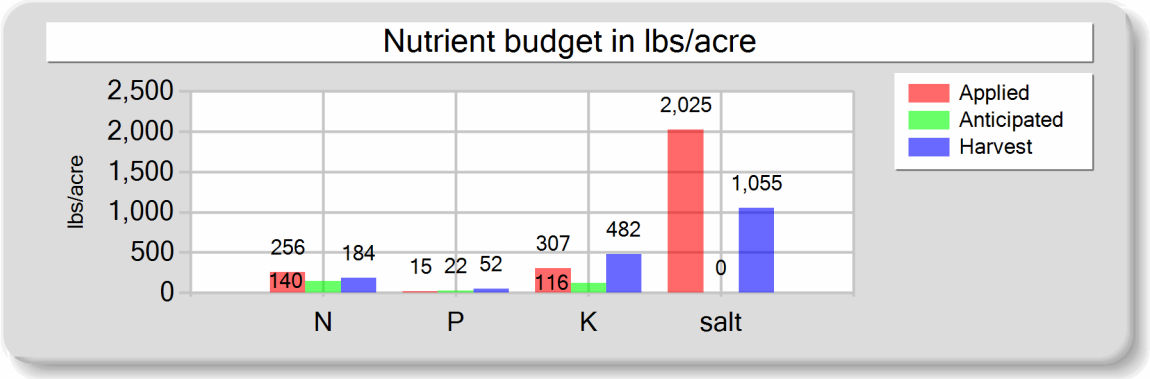
Fresh water applied	
28,320,000.00	*gallons*
1,042.93	*acre-inches*
29.80	*inches/acre*

Process wastewater applied	
882,000.00	*gallons*
32.48	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/25/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	245.48	15.00	307.03	1,760.86
Fresh water	3.26	0.00	0.00	264.37
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.75	15.00	307.03	2,025.23
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	183.67	51.68	481.80	1,055.29
Nutrient balance	72.08	-36.68	-174.77	969.94
Applied to removed ratio	1.39	0.29	0.64	1.92

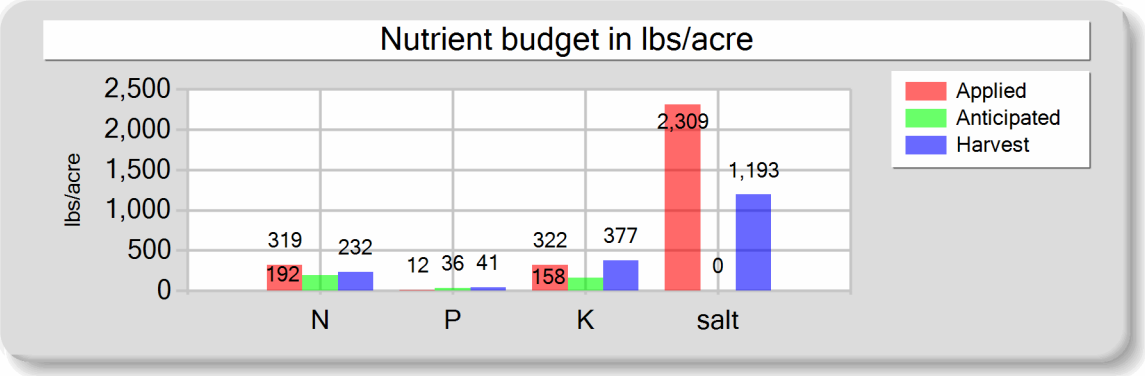
Fresh water applied	
2,400,000.00	*gallons*
88.38	*acre-inches*
8.84	*inches/acre*

Process wastewater applied	
336,000.00	*gallons*
12.37	*acre-inches*
1.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/18/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	301.07	11.96	322.10	1,420.19
Fresh water	10.98	0.00	0.00	888.94
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	319.05	11.96	322.10	2,309.13
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	231.84	40.88	377.44	1,193.19
Nutrient balance	87.21	-28.92	-55.34	1,115.94
Applied to removed ratio	1.38	0.29	0.85	1.94

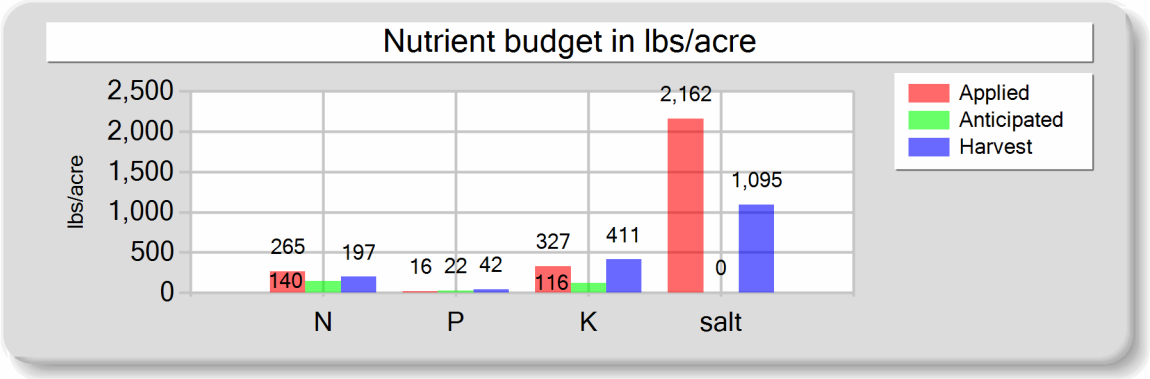
Fresh water applied	
8,070,000.00	*gallons*
297.19	*acre-inches*
29.72	*inches/acre*

Process wastewater applied	
336,000.00	*gallons*
12.37	*acre-inches*
1.24	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 11/25/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	254.70	16.01	326.64	1,857.69
Fresh water	3.76	0.00	0.00	304.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.46	16.01	326.64	2,162.40
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	196.80	42.27	411.49	1,095.19
Nutrient balance	68.66	-26.26	-84.85	1,067.21
Applied to removed ratio	1.35	0.38	0.79	1.97

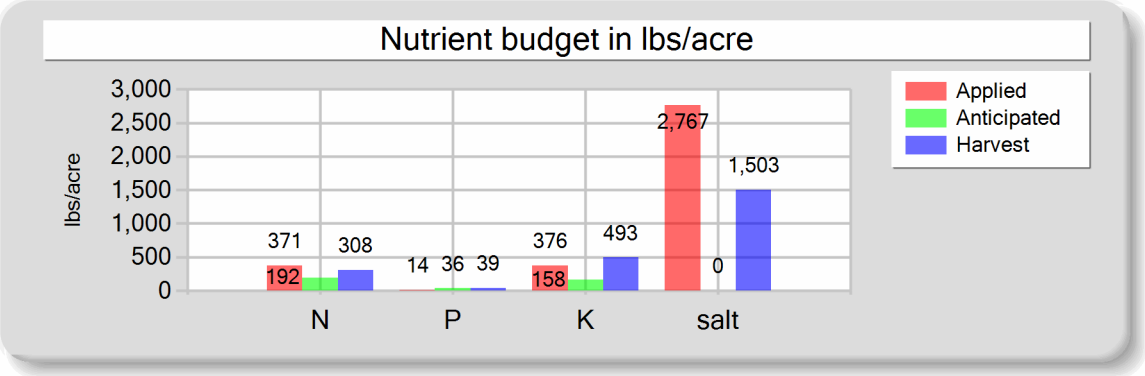
Fresh water applied	
21,300,000.00	*gallons*
784.41	*acre-inches*
10.19	*inches/acre*

Process wastewater applied	
2,772,000.00	*gallons*
102.08	*acre-inches*
1.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/18/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 05/18/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	349.58	13.69	375.80	1,610.25
Fresh water	14.29	0.00	0.00	1,157.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	370.87	13.69	375.80	2,767.30
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	308.22	39.03	493.04	1,503.10
Nutrient balance	62.64	-25.34	-117.25	1,264.20
Applied to removed ratio	1.20	0.35	0.76	1.84

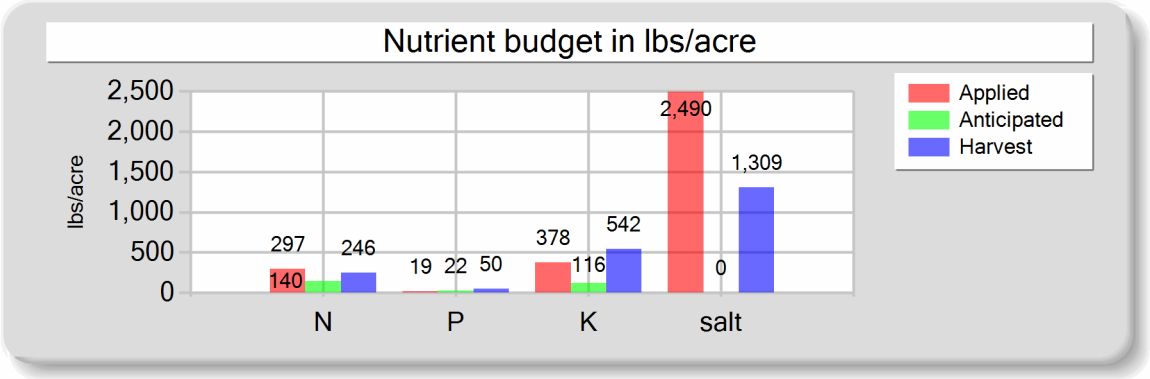
Fresh water applied	
80,880,000.00	*gallons*
2,978.54	*acre-inches*
38.68	*inches/acre*

Process wastewater applied	
3,024,000.00	*gallons*
111.36	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 11/25/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	285.51	18.58	377.83	2,126.77
Fresh water	4.49	0.00	0.00	363.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.00	18.58	377.83	2,490.28
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	245.94	49.98	542.44	1,308.67
Nutrient balance	51.05	-31.40	-164.61	1,181.61
Applied to removed ratio	1.21	0.37	0.70	1.90

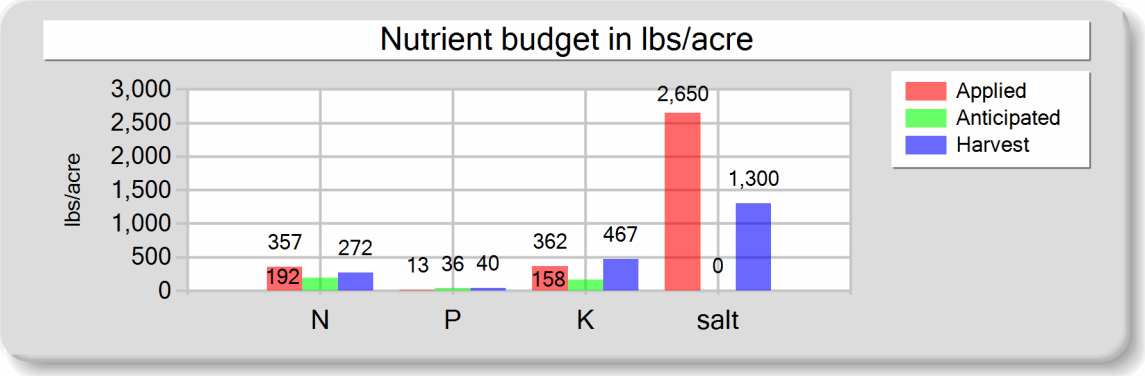
Fresh water applied	
3,300,000.00	*gallons*
121.53	*acre-inches*
12.15	*inches/acre*

Process wastewater applied	
420,000.00	*gallons*
15.47	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 05/19/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 05/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	336.47	13.17	361.71	1,549.87
Fresh water	13.59	0.00	0.00	1,100.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	357.06	13.17	361.71	2,650.31
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	271.80	39.60	467.40	1,299.60
Nutrient balance	85.26	-26.43	-105.69	1,350.71
Applied to removed ratio	1.31	0.33	0.77	2.04

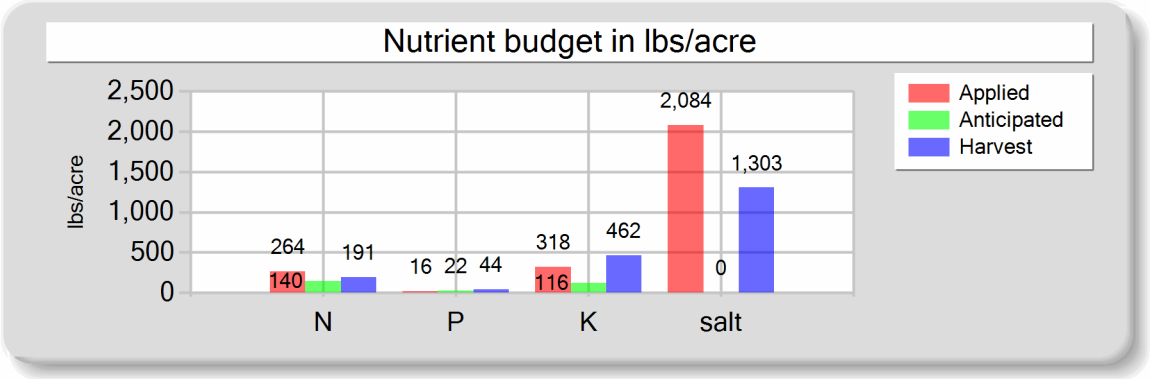
Fresh water applied	
9,990,000.00	*gallons*
367.90	*acre-inches*
36.79	*inches/acre*

Process wastewater applied	
378,000.00	*gallons*
13.92	*acre-inches*
1.39	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 11/25/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 11/25/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	253.95	15.52	317.62	1,821.58
Fresh water	3.24	0.00	0.00	262.09
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	264.18	15.52	317.62	2,083.67
Anticipated crop nutrient removal	140.00	22.40	116.20	0.00
Actual crop nutrient removal	190.70	44.21	461.57	1,303.08
Nutrient balance	73.48	-28.70	-143.95	780.59
Applied to removed ratio	1.39	0.35	0.69	1.60

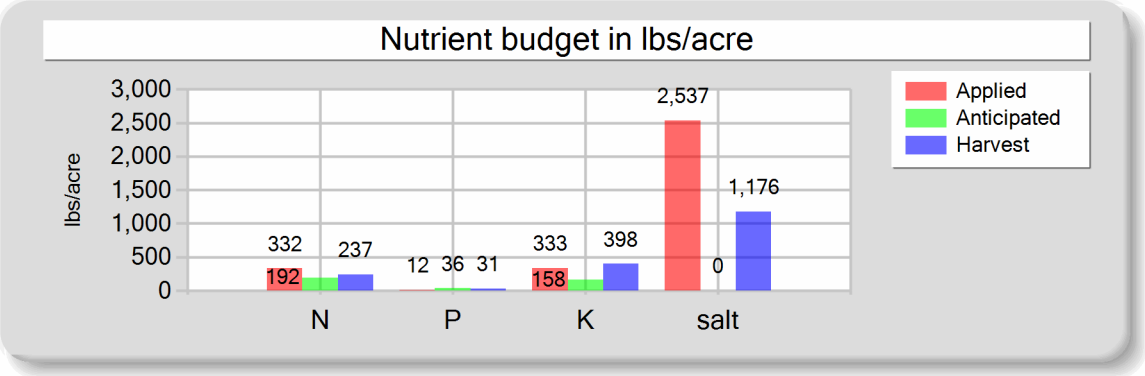
Fresh water applied	
27,600,000.00	*gallons*
1,016.41	*acre-inches*
8.76	*inches/acre*

Process wastewater applied	
4,032,000.00	*gallons*
148.48	*acre-inches*
1.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 05/19/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	311.45	12.37	333.21	1,469.16
Fresh water	13.19	0.00	0.00	1,068.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	331.64	12.37	333.21	2,537.46
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	237.27	30.63	397.72	1,176.23
Nutrient balance	94.38	-18.26	-64.51	1,361.23
Applied to removed ratio	1.40	0.40	0.84	2.16

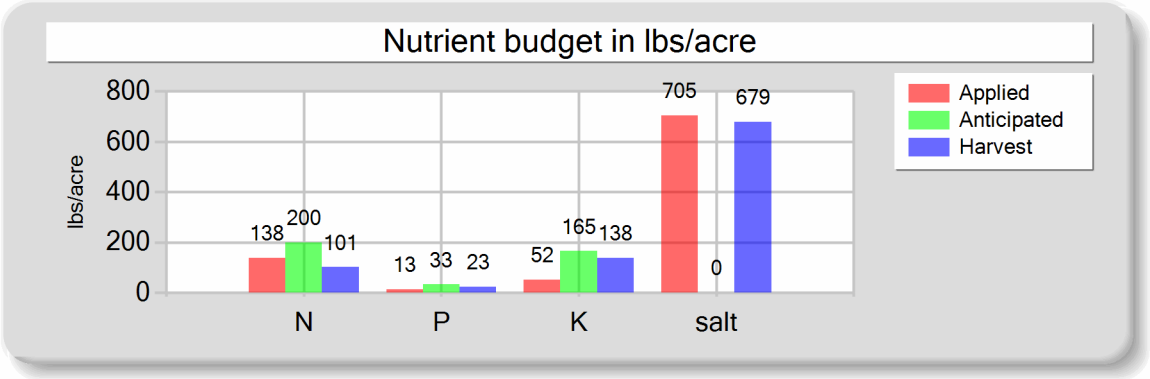
Fresh water applied	
112,500,000.00	*gallons*
4,142.99	*acre-inches*
35.72	*inches/acre*

Process wastewater applied	
4,032,000.00	*gallons*
148.48	*acre-inches*
1.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 11/26/2022: Oats, hay

Field name: Field 6 Crop: Oats, hay Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	129.13	12.71	51.65	571.08
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.58	0.00	0.00	133.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	137.70	12.71	51.65	704.60
Anticipated crop nutrient removal	200.00	32.50	165.00	0.00
Actual crop nutrient removal	101.02	22.81	137.87	678.85
Nutrient balance	36.68	-10.10	-86.22	25.76
Applied to removed ratio	1.36	0.56	0.37	1.04

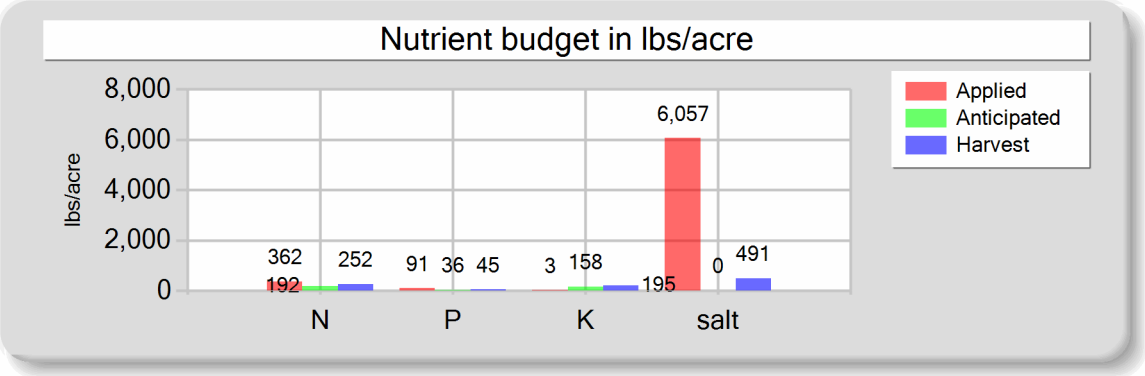
Fresh water applied	
3,750,000.00	*gallons*
138.10	*acre-inches*
4.60	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 06/29/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	264.32	91.19	2.64	5,127.81
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	10.96	0.00	0.00	929.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	362.28	91.19	2.64	6,057.11
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	251.72	44.93	194.88	490.56
Nutrient balance	110.56	46.26	-192.24	5,566.55
Applied to removed ratio	1.44	2.03	0.01	12.35

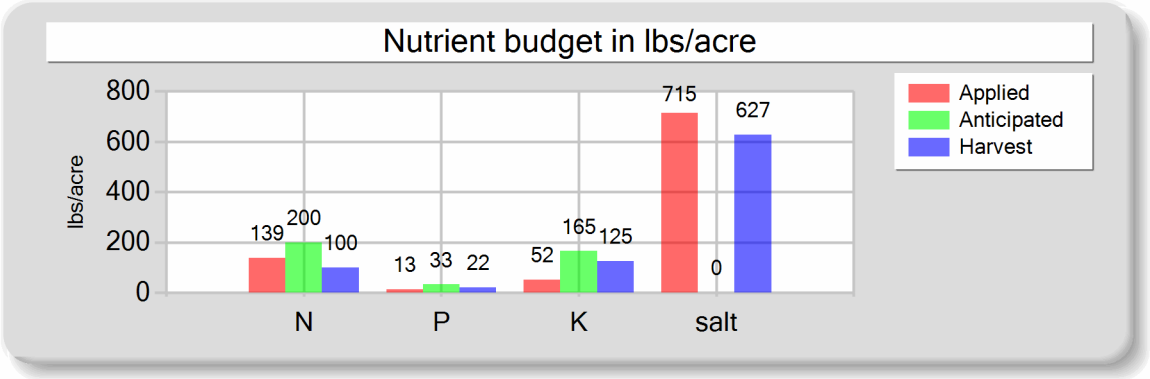
Fresh water applied	
26,100,000.00	*gallons*
961.17	*acre-inches*
32.04	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 11/26/2022: Oats, hay

Field name: Field 8 Crop: Oats, hay Plant date: 11/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	130.37	12.84	52.15	576.58
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.63	0.00	0.00	138.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	139.00	12.84	52.15	714.62
Anticipated crop nutrient removal	200.00	32.50	165.00	0.00
Actual crop nutrient removal	99.98	21.78	125.11	626.94
Nutrient balance	39.02	-8.94	-72.97	87.67
Applied to removed ratio	1.39	0.59	0.42	1.14

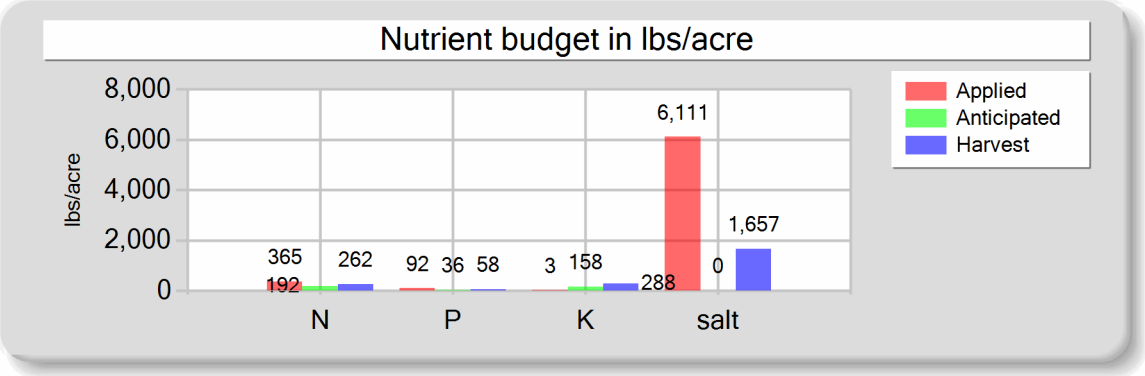
Fresh water applied	
16,800,000.00	*gallons*
618.69	*acre-inches*
4.76	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 06/29/2023: Corn, silage

Field name: Field 8 Crop: Corn, silage Plant date: 06/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	266.86	92.07	2.67	5,177.11
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	11.02	0.00	0.00	934.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	364.88	92.07	2.67	6,111.34
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	261.64	57.60	288.02	1,657.08
Nutrient balance	103.24	34.46	-285.35	4,454.27
Applied to removed ratio	1.39	1.60	0.01	3.69

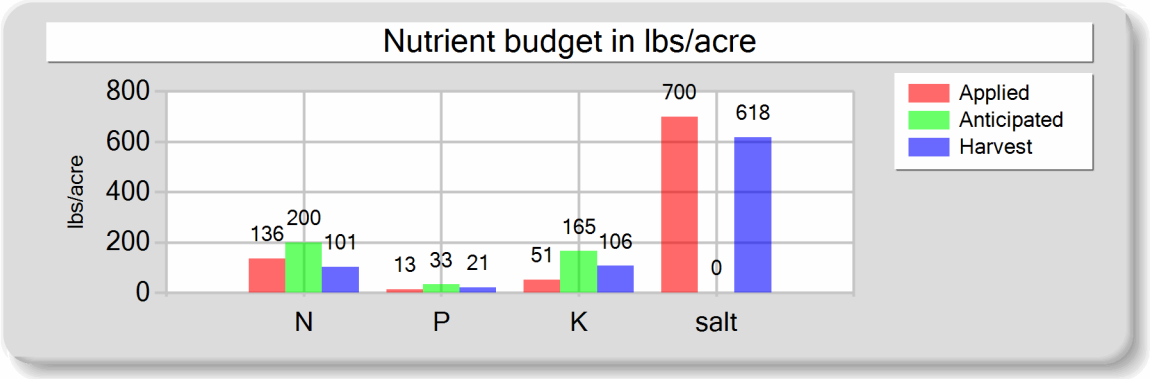
Fresh water applied	
113,700,000.00	*gallons*
4,187.18	*acre-inches*
32.21	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 9 - 11/27/2022: Oats, hay

Field name: Field 9 Crop: Oats, hay Plant date: 11/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	127.91	12.59	51.16	565.70
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	1.42	0.00	0.00	133.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	136.33	12.59	51.16	699.53
Anticipated crop nutrient removal	200.00	32.50	165.00	0.00
Actual crop nutrient removal	101.48	21.27	106.33	618.03
Nutrient balance	34.85	-8.67	-55.17	81.51
Applied to removed ratio	1.34	0.59	0.48	1.13

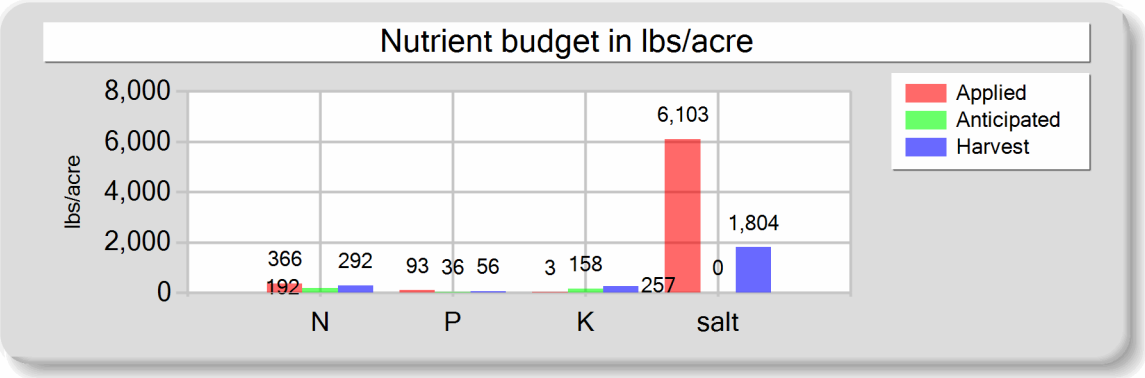
Fresh water applied	
20,400,000.00	*gallons*
751.26	*acre-inches*
4.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 9 - 06/30/2023: Corn, silage

Field name: Field 9 Crop: Corn, silage Plant date: 06/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	270.14	93.20	2.70	5,240.68
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	9.17	0.00	0.00	862.05
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	366.31	93.20	2.70	6,102.74
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	291.81	56.25	257.24	1,804.08
Nutrient balance	74.50	36.94	-254.54	4,298.66
Applied to removed ratio	1.26	1.66	0.01	3.38

Fresh water applied	
131,400,000.00	*gallons*
4,839.02	*acre-inches*
30.43	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator

Sample and source description: Separator

Sample date: 10/13/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,400.00	1,910.00	7,760.00							8.58
DL	100.00	100.00	100.00							0.01

Corral

Sample and source description: Corral

Sample date: 04/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,000.00	6,900.00	200.00							38.80
DL	100.00	100.00	100.00							0.01

Separator

Sample and source description: Separator

Sample date: 04/27/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,100.00	2,200.00	2,600.00							8.27
DL	100.00	100.00	100.00							0.01

Corral Manure

Sample and source description: Corral Manure

Sample date: 10/19/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	5,380.00	34,000.00							22.40
DL	100.00	100.00	100.00							0.01

Separator Manure

Sample and source description: Separator Manure

Sample date: 10/19/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,900.00	1,820.00	5,480.00							8.00
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

1st Qtr WW

Sample and source description: 1st Qtr WW

Sample date: 01/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	571.00	353.00	0.00	0.00	51.10	1,010.00								7,410.00	5,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2nd Qtr WW

Sample and source description: 2nd Qtr WW

Sample date: 05/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,180.00	532.00	0.00	0.00	55.90	1,180.00	221.00	150.00	164.00	5,420.00	0.00	255.00	264.00	8,410.00	7,340
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

3rd Qtr WW

Sample and source description: 3rd Qtr WW

Sample date: 07/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,010.00	585.00	0.00	0.00	34.70	1,130.00								7,800.00	3,700
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4th Qtr WW

Sample and source description: 4th Qtr WW

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,440.00	633.00	0.00	0.00	33.30	1,030.00								8,180.00	5,040
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW-Bledsoe PG&E

IW-Bledsoe PG&E

IW-Bledsoe PG&E

Sample description: IW-Bledsoe PG&E

Sample date: 01/19/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.33	0.00	1.20								145.00	125
DL	1.00	0.50	0.10								10.00	10

IW-Bledsoe TID

IW-Bledsoe TID

Sample description: IW-Bledsoe TID

Sample date: 01/19/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.51	0.00	1.20								137.00	128
DL	1.00	0.50	0.10								10.00	10

IW-Dairy TID

IW-Dairy TID

Sample description: IW-Dairy TID

Sample date: 01/19/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.63	0.00	1.30								142.00	132
DL	1.00	0.50	0.10								10.00	10

D. SOIL ANALYSES

Field 1

Field 1

Sample and source description: Field 1

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			55.00				
DL			4.00				

Field 2

Field 2

Sample and source description: Field 2

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			38.00				
DL			4.00				

Field 3

Field 3

Sample and source description: Field 3

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			55.00				
DL			4.00				

Field 5

Field 5

Field 5

Sample and source description: Field 5

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			26.00				
DL			4.00				

Field 6

6

Sample and source description: 6

Sample date: 10/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			30.00				
DL			4.00				

Field 8

8

Sample and source description: 8

Sample date: 10/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			55.00				
DL			4.00				

Field 9

Field 9

9

Sample and source description: 9

Sample date: 10/26/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			43.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Bledsoe Dryland - 11/27/2022: Oats, hay

Bledsoe Dryland

Sample and source description: Bledsoe Dryland

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 27.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,680.00	1,420.00	7,150.00		6.13
DL	100.00	100.00	100.00		0.01

Dairy Dryland - 11/25/2022: Oats, hay

Dairy Dryland

Sample and source description: Dairy Dryland

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,270.00	1,810.00	11,400.00		6.64
DL	100.00	100.00	100.00		0.01

Field 1 - 11/25/2022: Oats, silage-soft dough

Field 1

Sample and source description: Field 1

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,910.00	1,230.00	11,900.00		10.80
DL	100.00	100.00	100.00		0.01

Field 1 - 05/18/2023: Corn, silage

Field 1

Sample and source description: Field 1

Sample date: 09/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,510.00	480.00	4,800.00		5.18
DL	100.00	100.00	100.00		0.01

Field 2 - 11/25/2022: Oats, silage-soft dough

Field 2

Sample and source description: Field 2

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,290.00	1,770.00	16,500.00		13.00
DL	100.00	100.00	100.00		0.01

Field 2 - 05/18/2023: Corn, silage

Field 2

Sample and source description: Field 2

Sample date: 09/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,140.00	730.00	6,740.00		7.15
DL	100.00	100.00	100.00		0.01

Field 3 - 11/25/2022: Oats, silage-soft dough

Field 3

Sample and source description: Field 3

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,800.00	1,890.00	18,400.00		15.90
DL	100.00	100.00	100.00		0.01

Field 3

Sample and source description: Field 3

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,450.00	1,830.00	18,900.00		12.70
DL	100.00	100.00	100.00		0.01

Field 3 - 05/18/2023: Corn, silage

Field 3 - 05/18/2023: Corn, silage

Field 3

Sample and source description: Field 3

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,370.00	680.00	8,590.00		8.34
DL	100.00	100.00	100.00		0.01

Field 3

Sample and source description: Field 3

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,000.00	660.00	6,670.00		7.16
DL	100.00	100.00	100.00		0.01

Field 4 - 11/25/2022: Oats, silage-soft dough

Field 4

Sample and source description: Field 4

Sample date: 04/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,660.00	1,760.00	19,100.00		12.00
DL	100.00	100.00	100.00		0.01

Field 4 - 05/19/2023: Corn, silage

Field 4 - 05/19/2023: Corn, silage

Field 4

Sample and source description: Field 4

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,530.00	660.00	7,790.00		7.22
DL	100.00	100.00	100.00		0.01

Field 5 - 11/25/2022: Oats, silage-soft dough

Field 5

Sample and source description: Field 5

Sample date: 04/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,850.00	1,820.00	19,000.00		14.90
DL	100.00	100.00	100.00		0.01

Field 5

Sample and source description: Field 5

Sample date: 04/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,560.00	1,940.00	19,600.00		13.00
DL	100.00	100.00	100.00		0.01

Field 5 - 11/25/2022: Oats, silage-soft dough

Field 5

Sample and source description: Field 5

Sample date: 04/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 75.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,120.00	1,140.00	12,200.00		12.30
DL	100.00	100.00	100.00		0.01

Field 5 - 05/19/2023: Corn, silage

Field 5

Sample and source description: Field 5

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,710.00	630.00	7,650.00		6.91
DL	100.00	100.00	100.00		0.01

Field 5

Sample and source description: Field 5

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,880.00	630.00	8,180.00		7.68
DL	100.00	100.00	100.00		0.01

Field 5 - 05/19/2023: Corn, silage

Field 5

Sample and source description: Field 5

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,530.00	570.00	7,950.00		7.69
DL	100.00	100.00	100.00		0.01

Field 6 - 11/26/2022: Oats, hay

Field 6

Sample and source description: Field 6

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 19.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,060.00	1,820.00	11,000.00		6.72
DL	100.00	100.00	100.00		0.01

Field 6 - 06/29/2023: Corn, silage

Field 6

Sample and source description: Field 6

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,650.00	830.00	3,600.00		2.30
DL	100.00	100.00	100.00		0.01

Field 8 - 11/26/2022: Oats, hay

Field 8A

Sample and source description: Field 8A

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,630.00	1,880.00	10,800.00		6.04
DL	100.00	100.00	100.00		0.01

Field 8B

Sample and source description: Field 8B

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,880.00	1,830.00	9,990.00		7.37
DL	100.00	100.00	100.00		0.01

Field 8 - 06/29/2023: Corn, silage

Field 8

Sample and source description: Field 8

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,770.00	810.00	5,370.00		8.70
DL	100.00	100.00	100.00		0.01

Field 8 - 06/29/2023: Corn, silage

Field 8

Sample and source description: Field 8

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,290.00	810.00	5,500.00		8.70
DL	100.00	100.00	100.00		0.01

Field 8

Sample and source description: Field 8

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,860.00	1,070.00	5,350.00		9.00
DL	100.00	100.00	100.00		0.01

Field 9 - 11/27/2022: Oats, hay

Field 9A

Sample and source description: Field 9A

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,210.00	1,930.00	9,650.00		6.15
DL	100.00	100.00	100.00		0.01

Field 9 - 11/27/2022: Oats, hay

Field 9B

Sample and source description: Field 9B

Sample date: 06/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,340.00	1,830.00	10,900.00		7.50
DL	100.00	100.00	100.00		0.01

Field 9 - 06/30/2023: Corn, silage

Field 9

Sample and source description: Field 9

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,980.00	960.00	4,390.00		8.60
DL	100.00	100.00	100.00		0.01

Field 9

Sample and source description: Field 9

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,790.00	810.00	3,100.00		7.80
DL	100.00	100.00	100.00		0.01

Field 9 - 06/30/2023: Corn, silage

Field 9

Sample and source description: Field 9

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	670.00	2,650.00		2.10
DL	100.00	100.00	100.00		0.01

Field 9

Sample and source description: Field 9

Sample date: 10/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,650.00	780.00	2,690.00		1.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

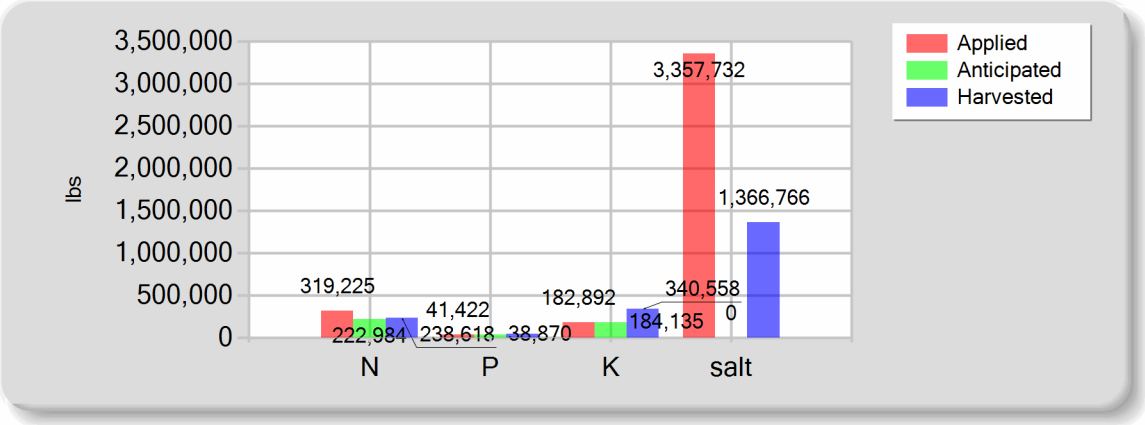
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

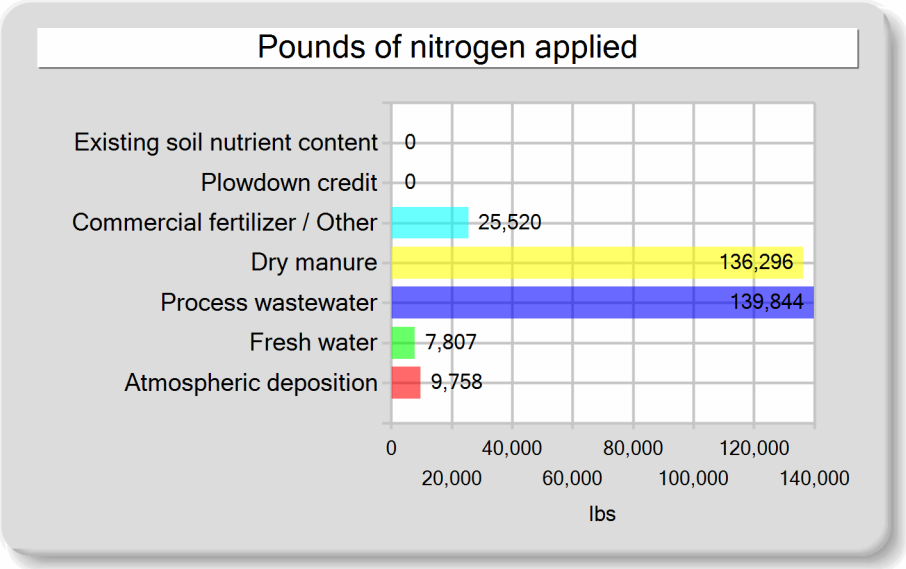
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

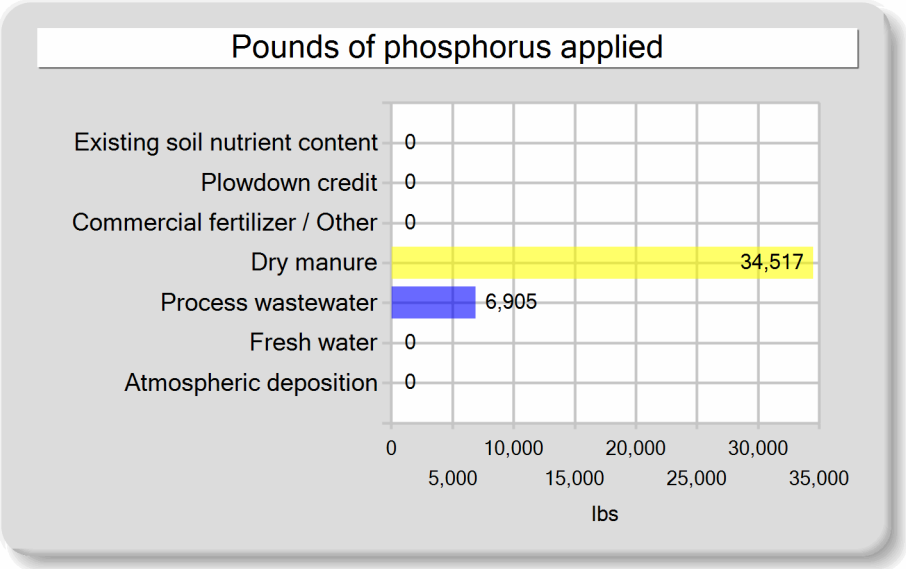
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	25,520.00	0.00	0.00	0.00
Dry manure	136,296.06	34,516.70	21,144.72	1,884,457.09
Process wastewater	139,844.11	6,905.49	161,747.63	811,468.47
Fresh water	7,807.29	0.00	0.00	661,806.23
Atmospheric deposition	9,758.00	0.00	0.00	0.00
Total nutrients applied	319,225.46	41,422.20	182,892.35	3,357,731.79
Anticipated crop nutrient removal	222,984.00	38,869.70	184,135.40	0.00
Actual crop nutrient removal	238,618.32	45,191.47	340,558.44	1,366,766.02
Nutrient balance	80,607.14	-3,769.27	-157,666.09	1,990,965.76
Applied to removed ratio	1.34	0.92	0.54	2.46

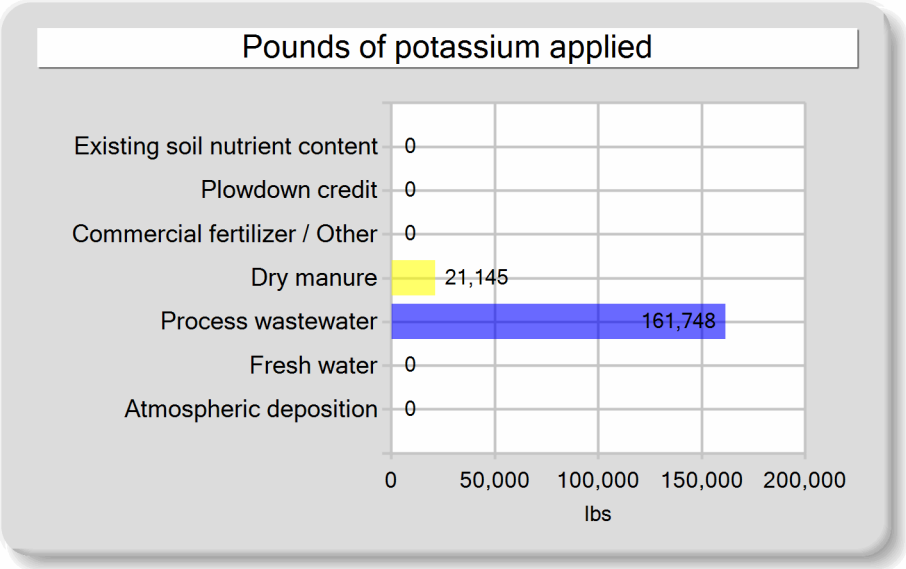
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

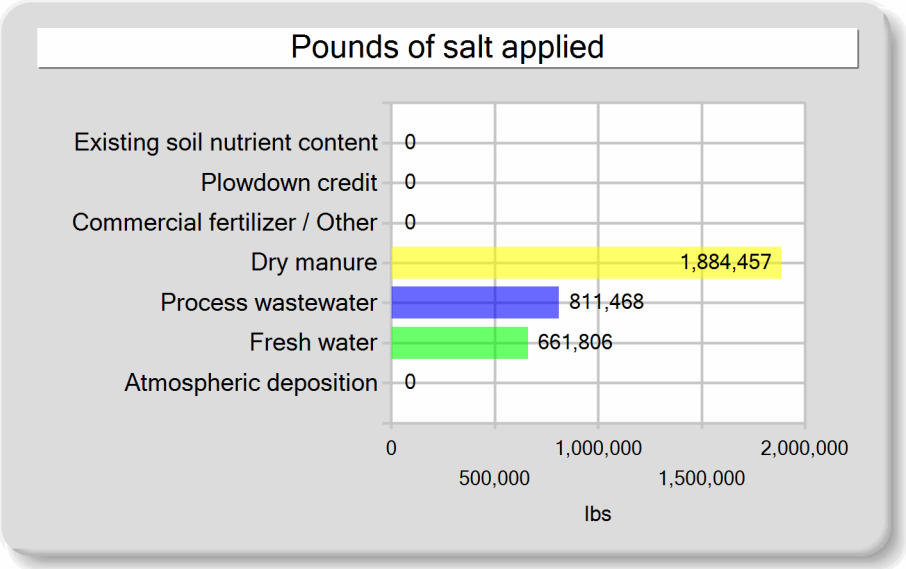


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Due to an oversight, well water samples were not procured during the 2023 reporting period. Irrigation Well samples from 2022 were used to complete the Annual Report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Vance W Ahlem (Jun 28, 2024 08:05 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Vance William Ahlem	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 28, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

2023 Countyline Annual Report

Final Audit Report 2024-06-28

Created:	2024-06-28
By:	Sean Locke (slocke@fragservices.com)
Status:	Signed
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Signer vahlem@countylinemilk.com entered name at signing as Vance W Ahlem
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